



Quantitative Data Analysis

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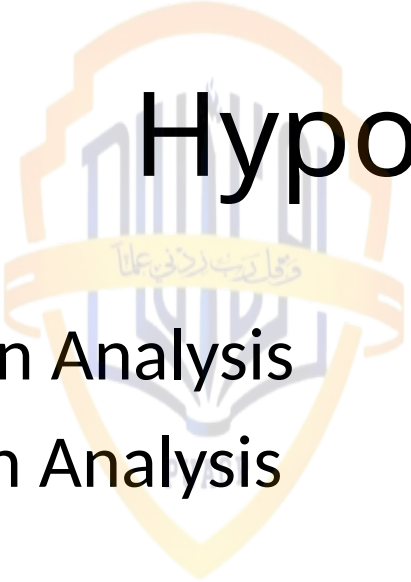


Data Refinement

1. Download Excel Sheet from
2. Change reverse-worded items
3. Download SPSS Software
4. Copy data from the excel sheet to SPSS data view
5. Create variables in the variable view in SPSS
6. Recode into same variable:
 1. Strongly Disagree → 1
 2. Disagree → 2
 3. Neutral → 3
 4. Agree → 4
 5. Strongly Agree → 5
7. Assign codes to demographic variables

Preliminary Data Analysis

1. What to do with blank responses?
2. Add one ID number and compute box plotting to identify outliers
3. Reliability analysis
4. Compute variables
5. Descriptive Analysis
 1. Mean
 2. S.D
 3. Ranges
 4. Frequency in percentage
6. Demographic Analysis
7. Independent Sample T Test
8. One-way ANOVA Test

The logo of the University of Al-Qadisiyah is a shield-shaped emblem. It features a stylized building facade with a central archway and a banner across the middle containing Arabic text. The colors are primarily orange, yellow, and blue.

Hypotheses Testing

1. Correlation Analysis
2. Regression Analysis



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Types of Analysis

DATA
ANALYSIS

Univariate

Bivariate

Multivariate



Univariate Analysis

- **Definition:** Univariate analysis focuses on a single variable or attribute at a time. It is used to describe and understand the characteristics and distribution of that variable.
- **Methods:** Measures of central tendency (mean, median, mode), measures of dispersion (variance, standard deviation), frequency distributions, and histograms are commonly used in univariate analysis.



Bivariate Analysis

- **Definition:** Bivariate analysis involves the examination of the relationship between two variables. It helps understand how one variable relates to or influences another.
- **Methods:** Scatter plots, correlation analysis, chi-squared tests, and t-tests are often used in bivariate analysis.

Multivariate Analysis

The logo of the University of Al-Qadisiyah is positioned behind the title. It features a shield-like emblem with a yellow border. Inside, there is a stylized blue and white building or book structure. A yellow banner across the middle contains the university's name in Arabic script.

- **Definition:** Multivariate analysis deals with the analysis of more than two variables simultaneously. It explores complex relationships and patterns among multiple variables.
- **Methods:** Principal Component Analysis (PCA), Factor Analysis, Multiple Regression Analysis, Cluster Analysis, and Discriminant Analysis are examples of multivariate analysis techniques.

Major Analysis Techniques



DATA ANALYSIS

Descriptive

Factor

Inferential

Descriptive Analysis

- **Purpose:** Descriptive analysis aims to summarize and describe the main features of a dataset. It provides a clear and concise overview of the data.
- **Methods:** Measures of central tendency (mean, median, mode), measures of dispersion (variance, standard deviation), frequency distributions, histograms, and summary statistics are commonly used in descriptive analysis.



Factor Analysis

- **Purpose:** Factor analysis is a statistical technique used in data analysis to identify underlying factors or latent variables that can explain patterns of correlations among observed variables. It's commonly used in fields like psychology, sociology, economics, and marketing to reduce the dimensionality of data and understand the underlying structure or concepts that drive observed relationships.
- **Methods:** EFA, CFA etc.



Inferential Analysis

- **Purpose:** Inferential analysis involves making inferences or predictions about a population based on a sample of data. It helps draw conclusions and test hypotheses.
- **Methods:** Hypothesis testing, confidence intervals, regression analysis, analysis of variance (ANOVA), and chi-squared tests are common techniques in inferential analysis.



Final Step of Research!

1. Discussion on the Results
2. Conclusion
3. Limitations
4. Future Recommendations